26/10/2020

1. $X(z) = \dfrac{10z}{z^2 - 3z + 2}$ $= \dfrac{10z}{(z-1)(z-2)}$

$$10z\left[\frac{1}{(z-1)(z-2)}\right] = \frac{-1}{z-1} + \frac{1}{z-2}$$

$$= 10\left[\frac{-z}{z-1} + \frac{z}{z-2}\right]$$

Applying inverse.

$$10\left[-1 + 2^n\right].$$

2. $X(z) = \dfrac{z}{z^2 + 7z + 10}$

$$= z\left[\frac{1}{(z+5)(z+2)}\right]$$

Apply partial fractions

$$= \left[\frac{1/3\, z}{z+2} + \frac{-1/3\, z}{z+5}\right]$$

Applying inverse se

$$= 1/3(-2)^n - 1/3(-5)^n.$$

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3. $X(z) = \dfrac{8z^2}{(2z-1)(4z-1)}$.

$$\frac{X(z)}{8(z)} = \frac{z}{(2z-1)(4z-1)}$$

$$\frac{z}{(2z-1)(4z-1)} = \frac{A}{2z-1} + \frac{B}{4z-1}$$

$A = 1/2 \quad B = -1/2$

$$\frac{X(z)}{8(z)} = \frac{1/2}{2z-1} - \frac{1/2}{4z-1}$$

$$= \frac{4z}{2z-1} - \frac{4z}{4z-1}$$

$$= \frac{2z}{z-1/2} - \frac{z}{z-1/4}.$$

Applying Inverse

$$2(1/2)^n - (1/4)^n.$$

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4. $\dfrac{z^2+z}{(z-1)(z^2+1)}$

$$\frac{z^2}{(z-1)(z^2+1)} + \frac{z}{(z-1)(z^2+1)}$$

$$z^2\left[\frac{z/2}{z^2+1} - \frac{1/2}{z^2+1} + \frac{1/2}{z-1}\right] + \cancel{1/2}\,\cancel{\cos n}$$

$$+ z\left[\frac{z/2}{z^2+1} - \frac{1/2}{z^2+1} + \frac{1/2}{z-1}\right]$$

$$\frac{z}{2}\,\frac{z^2}{z^2+1} - \frac{1}{2}\,\frac{z^2}{z^2+1} + \frac{z}{2}\,\frac{z}{z-1}$$

$$= \frac{1}{2}\cos\frac{(n+1)\pi}{2} - \frac{1}{2}\cos\frac{n\pi}{2} + \frac{1}{2}(1)^{n+1}$$

$$z\left[\frac{z/2}{z^2+1} - \frac{1/2}{z^2+1} + \frac{1/2}{z-1}\right]$$

$$= 2\left[\frac{z^2}{z^2+1} - \frac{z}{z^2+1} + \frac{1}{2}\,\frac{z}{z-1}\right]$$

$$= 2\cos\frac{n\pi}{2} - 2\sin\frac{n\pi}{2} + \frac{1}{2}$$

5. $\dfrac{2z}{z^3-z^2+z-1}$

$$2z\left[\frac{z/2 - 1/2}{z^2+1} + \frac{1/2}{z-1}\right]$$

$$= 2z\left[\frac{z/2}{z^2+1} - \frac{1/2}{z^2+1} + \frac{1/2}{z-1}\right]$$

$$= 2\left[\frac{z^2}{z^2+1} - \frac{z}{z^2+1} + \frac{1}{2}\,\frac{z}{z-1}\right]$$

$$= 2\cos\frac{n\pi}{2} - 2\sin\frac{n\pi}{2} + 1.$$

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